

A. Familiar commons examples

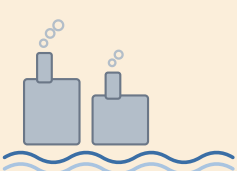
in human social dilemmas



Shepherds

share a pasture

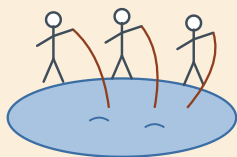
*overgrazing
depletes the field*



Factories

share a river

*pollution shifts
cost downstream*



Fisherman

share a pond

*overfishing cause
population collapse*

B. Artificial commons examples

in LLM agent systems



Agents

share a power grid



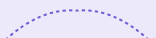
power consumption

*coordinated load shifts
destabilize the grid*



Agents

share a databases



memory store

*unverified writes
corrupt the state*



Agents

share critical codebases



*unverified merges
break reliant applications*

C. Design space

institutional functions × mechanisms

Institutional functions

what an institution must do

norms

monitoring

reputation

sanctions

constraints

adjudication

repair

Mechanism families

how it shapes behavior

normative

social

epistemic

incentive

constraint

restorative

POSITION

Multi-agent AI safety requires intentional institutional design. Even individually aligned agents can produce collectively unsafe outcomes when sharing finite resources. We believe agent systems should be studied as governed systems whose institutions shape collective behavior.

— our central claim

Contribution 1

Aligned individuals are not enough; institutions are needed.

Contribution 2

Distinguishing 7 functions from 6 mechanism families.

Contribution 3

An evaluation agenda for cooperation-shaping institutions.